Verification Plan

Project: DATERA EDU. Development of an intelligent system for monitoring students' knowledge acquisition using speech recognition and large language model technologies for analysing teacher-student dialogue

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FR stands for a functional requirement and NFR for a non-functional one. The numbers after them are the numbers of the project requirements.

Timeline: the MVP is built on ready-made models from November 2026 to early March 2027. The trial fine-tuning on open data takes place in late winter 2027. Stage 2, from March to April 2027, adds tests, weekly summaries, notifications and fine-tuning on the teachers' corrections. The pilot runs from March to April 2027.

1. Purpose

This document describes how it is shown that the project reaches its aim: which tests are run, what is measured, under which conditions, and what the results are compared with.

The aim of the project is simple to say. A teacher uploads course materials, then the recording of a one-to-one lesson. The system shows which topics the student answered correctly and which not, keeps this history in a knowledge map, and sends a report to the teacher, the student and the parent. The teacher no longer has to watch the whole lesson again. The final word stays with the teacher, who can confirm or change any rating (FR-8.7). The MVP covers one-to-one lessons in English (often with explanations in Russian) and in Russian. Kazakh and group lessons come after the MVP.

The MVP is accepted when two things are true. The end-to-end scenario "upload a recording, get reports for three roles, see the knowledge map" passes on real recordings of one-to-one lessons, and the NFR-5 targets are reached on the test set. Stage 2 is accepted by its test scenarios and by the report that compares the models (FR-16.5).

2. How it is shown that the aim is reached

The aim is split into seven claims. Each one has a test, a metric, a target and a baseline, and the aim is reached when all seven hold.

#	Claim and test	Main metrics and target	Baseline
C1	The system turns a real lesson in English, Russian or mixed speech into a correct transcript with speakers. Test: ML pipeline on the test set	WER up to 15 % English, 20 % Russian, 25 % mixed; DER up to 20 %; teacher found in at least 95 % of recordings (NFR-5.1, NFR-5.3, NFR-5.4)	Same pipeline without cleaning; smaller Whisper model

#	Claim and test	Main metrics and target	Baseline
C2	The topic statuses match the teacher's own rating. Test: control set of 5 to 10 lessons rated by a teacher (FR-16.2)	Status agreement at least 70 % (NFR-5.5); valid JSON at least 95 % (NFR-5.6)	Majority class; the same analysis on a checked transcript
C3	The mastery index is calculated in a way that fits the teacher's judgment. Test: index variants compared on the control set	Agreement with the teacher for each variant; the best one goes into the system	Three-level index of the MVP (D = 0, 50 or 100 %)
C4	Reports are ready fast enough for daily work. Test: timing and load tests	Report in 30 min or less for 60 min of audio; 10 recordings a day; API p95 up to 500 ms; page up to 3 s with 30 users (NFR-1.1 to NFR-1.4)	Manual review, which takes about as long as the lesson
C5	Every MVP function works safely for every role. Test: end-to-end scenarios, functional checklist, security tests	9 of 9 MVP scenarios; 100 % of MVP FR; 0 access violations	None
C6	Teachers trust the ratings, and the system saves their time. Test: pilot with 2 to 3 teachers for 2 to 4 weeks	At least 70 % of statuses left unchanged (NFR-5.8); proposed: teacher time down by 50 % or more, SUS 68 or more	Current manual process, measured before the pilot
C7	Fine-tuning gives the project its own analysis model that beats the same model without it (stage 2). Test: base, fine-tuned and cloud models on the control set; trial on open data first	Fine-tuned better than base (NFR-5.7); connected if no more than 5 p.p. below the cloud model	Base open model; cloud LLM of the MVP

The targets of C1, C2, C4, C5 and C7 are project requirements. In C6 only NFR-5.8 is a requirement. The requirements ask for nothing more than a satisfaction questionnaire in the pilot, so the time and SUS targets are proposed by the team, and the supervisor has to approve them. C3 has no fixed number: index variants are compared and the best one is kept.

3. Test levels

Level	What is tested	Tools	Who and when
Unit tests	Go API logic (index formula, statuses, access rules, consent check); ML service helpers; frontend logic (status and color rules, API client, forms)	Go testing, pytest, Vitest + React Testing Library	Each developer for own code; every commit (NFR-7.3)
Contract tests	The typed frontend client is generated from the OpenAPI file; the build fails if the frontend uses a field the API does not have	openapi-typescript, TypeScript compiler	Yermakhan; every commit and every change of the OpenAPI file

Level	What is tested	Tools	Who and when
Integration tests	API + PostgreSQL + RabbitMQ + MinIO in Docker Compose; API and ML service through the queue (materials, lesson and report jobs)	Go tests with test containers, pytest	Tairkhan, David; every merge to main
ML evaluation	WER by language, DER, effect of cleaning, teacher detection, LLM quality, effect of recognition errors on the analysis; control WER on Kazakh	Python scripts: jiwer, pyannote.metrics	David; research stage and after every change of a model or prompt
Topic extraction check	How many topics proposed by the LLM from the materials the teacher keeps without changes (FR-3.5, FR-3.6)	Logs of topic edits	David; research stage and pilot
Fine-tuning experiments	Trial: fine-tuned against base model on open data (FR-16.0). Stage 2: base, fine-tuned and cloud model on the control set (NFR-5.7, FR-16.5)	Hugging Face Transformers, PEFT, vLLM	David; late winter 2027, then early spring 2027
Functional tests	All MVP requirements by checklist	Checklist + Playwright	Whole team; before each demo and before the pilot
UI end-to-end tests	Scenarios S1 to S9 through the real interface in all three cabinets (section 7.1)	Playwright (Chromium, Firefox, WebKit)	Yermakhan, with test data from the backend; every merge on the mock API, before each demo on the real API
Screen and browser tests	Every page at 360, 768 and 1440 px; two last versions of Chrome, also Edge, Firefox, Safari (NFR-6.1, NFR-6.2)	Playwright viewports and screenshots, a real phone	Yermakhan; integration and before the pilot
Accessibility and usability	Keyboard, contrast, labels; upload and open a report in 3 steps or less (NFR-6.5); no special terms for parents (NFR-6.4)	axe-core, Lighthouse, click-count checklist	Yermakhan; integration
Performance tests	Processing time, 10 recordings in a row, API time with 30 users, page load (NFR-1)	k6, Lighthouse, timing logs	Tairkhan; page load: Yermakhan; integration
Reliability tests	No lost jobs, resume, retries, daily backup, availability (NFR-2)	Stop services during processing, restore from backup, uptime monitor	Tairkhan; integration

Level	What is tested	Tools	Who and when
Security tests	Access between roles and organizations, signed links, login lock, OWASP Top 10 (NFR-3)	Manual tests, API requests with wrong tokens, OWASP ZAP baseline scan	Tairkhan, Yermakhan; integration and before the pilot
Pilot	Real work of 2 to 3 teachers for 2 to 4 weeks	Questionnaire, time diary, logs of “Confirm” and “Change”	David; March to April 2027

Unit tests must cover at least 50 % of the backend business logic (NFR-7.4), and CI checks this with go test -cover. There is no such number for the frontend, so a separate target is set: at least 50 % for logic code (utilities, hooks, API client), checked with vitest --coverage. Page layout is covered by end-to-end tests instead.

4. Metrics

4.1 ML quality (C1, C2, C3, C7)

Metric	How it is calculated	Target	Requirement
WER by language	WER = (S + D + I) / N: S substituted, D deleted, I inserted words, N words in the reference. Library jiwer. Separate for English, Russian and mixed lessons	Up to 15 % / 20 % / 25 % (mixed is preliminary)	NFR-5.1
WER on Kazakh (control)	Same formula on 3 to 5 Kazakh lessons	Reported only, not part of acceptance	Research part
Effect of cleaning (ΔWER)	WER without cleaning minus WER with cleaning, on noisy recordings	More than 0	NFR-5.2
DER	(missed speech + false alarm + speaker confusion) / total speech time. Library pyannote.metrics, collar 0.25 s	Up to 20 % (preliminary)	NFR-5.3
Teacher detection	Recordings where the teacher is found correctly / all recordings	At least 95 %	NFR-5.4
Topic status agreement	Topics where the system’s status (red, yellow, green) equals the teacher’s status / topics rated by both, on the control set	At least 70 %	NFR-5.5
Weighted Cohen’s kappa	Agreement corrected for chance, linear weights for red < yellow < green	Reported; κ above 0.4 is wanted (proposed)	Supports NFR-5.5
Effect of recognition errors	Status agreement when the LLM reads the checked transcript minus agreement on the recognized one, same lessons	Reported	Research part

Metric	How it is calculated	Target	Requirement
Index variants	Agreement with the teacher for each index variant (three levels or a 0 to 100 % scale, different weights)	Reported; the best variant goes into the system	FR-8
Topics outside the materials	Fragments marked “outside the materials” that the teacher also sees as outside, checked by hand	Reported	FR-7.8
Valid JSON	LLM answers that pass the JSON schema on the first try / all answers	At least 95 %	NFR-5.6
Fine-tuning effect	Agreement of the fine-tuned model minus agreement of the same model without fine-tuning, on the control set	More than 0	NFR-5.7
Gap to the cloud model	Agreement of the cloud model minus agreement of the fine-tuned model	Up to 5 p.p. to connect the fine-tuned model	FR-16.6
Names in LLM requests	Real names from the participant list found in the text sent to the LLM	0	FR-7.6, NFR-4.2

Before WER is counted, both texts go through the same normalization. Everything is lower case, punctuation is removed, “ё” becomes “е”, and numbers are written as words. Filler words (“эм”, “ну”, “um”, “uh”) stay. English short forms are written one way (“don’t”, never “do not”). In mixed lessons each word is compared in its own language and script, so a Russian word written in Latin letters is an error.

4.2 Speed and reliability (C4)

Metric	Definition	Target	Requirement
Time to report	From the end of upload to “Ready”, 60-minute lesson, empty queue	30 min or less	NFR-1.1
Time per stage	Preparation, cleaning, recognition, diarization, LLM analysis, index, LLM reports, from logs with one lesson ID (NFR-7.5)	Reported in the thesis	Research part
Real-time factor	Processing time / audio length, per stage	Reported	None
Throughput	10 recordings uploaded one after another: all processed in order, none lost	10 of 10 within 24 h	NFR-1.2
API response time	95th percentile for lists, cards and reports, 30 virtual users	500 ms or less	NFR-1.3
Page load time	Cabinet page, network slowed to 10 Mbit/s, 30 users at once	3 s or less	NFR-1.4
No lost jobs	ML service and RabbitMQ restarted during processing	0 lost jobs	NFR-2.1
Resume	After a restart, processing goes on from the last finished stage	Yes, in 100 % of tries	NFR-2.2

Metric	Definition	Target	Requirement
Retries	A stage fails on purpose: 3 retries, then “Error” with a reason	Yes	NFR-2.3
Backup	Daily database backup kept for 7 days; one restore is tested	Restore works	NFR-2.4
Availability	Time the cabinets work during the pilot (uptime monitor)	At least 95 % a month	NFR-2.5

4.3 Trust and practical value (C6)

Metric	How it is measured	Target
Statuses left unchanged	Topic statuses the teacher confirmed or left as they were / all statuses in the pilot	At least 70 % (NFR-5.8)
Teacher time per lesson	Time diary: minutes to understand what the student did not learn and to prepare the next lesson. One week before the pilot (manual) and during the pilot (with reports)	Proposed: 50 % less or better
Usefulness of reports	Questionnaire for teachers and parents, 5-point scale: “the report is correct”, “the report is clear”, “I would use it every week”	Proposed: average 4 of 5 or more
Usability	System Usability Scale (10 standard questions) for each cabinet	Proposed: SUS 68 or more
Parent understanding	Parents say in their own words what the child must repeat; this is compared with the report	Proposed: 80 % correct or more

NFR-5.8 and the time diary come first for a reason. In an interview, an English teacher said that she would stop using a system that often makes mistakes or takes a lot of time to check.

4.4 Frontend quality (C4, C5, C6)

Metric	How it is measured	Target	Requirement
Page load (LCP)	Lighthouse on the production build, 10 Mbit/s, empty cache; 5 pages (teacher home, student list, lesson report, student home, parent home); median of 3 runs	3 s or less	NFR-1.4
JavaScript size	JavaScript of the first page, gzip, from the Vite build report; each cabinet is a separate chunk	Proposed: 300 KB or less	Supports NFR-1.4
Steps to main actions	Clicks from the cabinet home page to upload a recording and to open a report	3 or less	NFR-6.5
Phone layout	No horizontal scroll, no cut or overlapping text, buttons at least 40 px high; screenshots at 360 × 800 for every page, a real phone for the main pages	0 problems	NFR-6.2

Metric	How it is measured	Target	Requirement
Browsers	End-to-end tests in the two last versions of Chrome; Edge, Firefox and Safari are checked	100 % in Chrome; problems in others are listed and fixed if possible	NFR-6.1
Texts in translation files	Texts written straight in the code, found by the ESLint rule i18next/no-literal-string	0	NFR-6.3
No special terms for parents	Words like “index”, “diarization”, “transcript”, “LLM”, “WER” on parent pages; the same word list goes into the prompt for the parent report (with David)	0 words	NFR-6.4, FR-13.2
Time-code accuracy	Difference between the time code in a report and the player position after a click; 10 time codes of a test lesson	1 s or less	FR-9.4
Status on screen	Time between a stage change on the server and its display, without reload	Proposed: 10 s or less (polling every 5 s)	FR-4.5
Upload of a recording	A 2 GB file uploads with progress; a wrong format or size is stopped before upload; a broken connection shows an error and “Retry”	Works in 100 % of tries	FR-4.1 to FR-4.3
Upload of materials	PDF, DOCX, PPTX, TXT and MD up to 20 MB upload; other formats and bigger files are stopped with a clear message; the topic list appears for confirmation	Works in 100 % of tries	FR-3.4 to FR-3.6
“Confirm” and “Change”	After a change the report, the index and the knowledge map show the new value without reload, and the change is in the history	Yes	FR-8.7, FR-8.6
Long session	A page stays open more than 15 minutes and requests still work without a new login	Yes	NFR-3.3
Accessibility	axe-core on every page; all main actions by keyboard; text contrast at least 4.5 : 1 (WCAG 2.1 AA)	Proposed: 0 critical problems, Lighthouse accessibility 90 or more	Supports C6
Match with the mockups	Sizes, spacing, colors and texts of a page against its Figma frame, at 1440 px and 360 px	All big differences fixed or written down	Approved mockups

Targets marked “proposed” are set by the team, and the supervisor has to approve them.

5. Baselines

A number on its own says little. For each claim a baseline is also measured under the same conditions.

Claim	Baseline	Why this one	What has to be shown
C1, recognition	B1: same pipeline without cleaning (large-v3 on raw audio)	Shows the effect of cleaning (NFR-5.2)	WER with cleaning is lower than B1 on noisy recordings
C1, recognition	B2: smaller model (large-v3-turbo)	Shows the trade-off between quality and speed	large-v3 has lower WER; the time difference is reported
C1, diarization	B3: pyannote.audio with default settings, no hint about the number of speakers	Shows the effect of tuning (two speakers in a one-to-one lesson)	DER of the final setup is not higher than B3
C2	B4: majority class, every topic gets the most common teacher status	70 % can be easy if most topics have the same status	Agreement and kappa clearly higher than B4
C2	B5: the same LLM analysis on the checked transcript	Upper bound: shows how much recognition errors cost	System agreement is close to B5; the gap is reported
C2	B6: 2 to 3 cloud LLMs on the same control lessons	The model is chosen by data	The chosen model has the best agreement and JSON rate
C3	B7: three-level index of the MVP (I = D, D = 0, 50 or 100 %)	Starting version of the index (FR-8)	Another variant is taken only if it agrees with the teacher better
C4, C6	B8: current manual process: the teacher listens to the recording again or relies on memory	This is what the system replaces	Report time and teacher time are lower than B8
C6, interface	B9: usability test of the clickable Figma prototype before coding: 3 to 5 users (teachers and parents), 3 tasks each (“upload a lesson”, “find what the child must repeat”, “confirm or change a topic status”)	Shows if the design is clear before code is written	Task success and SUS in the pilot are not lower than in B9
C7	B10: the same open model without fine-tuning, and the cloud LLM of the MVP, on the same control set	It has to be shown that fine-tuning helps (NFR-5.7) and three models are compared (FR-16.5)	Fine-tuned is better than the base model; the gap to the cloud model is reported

6. Measurement conditions

6.1 Dataset

The test data must meet these conditions:

- At least 20 recordings of real one-to-one lessons in 2 to 3 subjects. Every participant gives consent, and for students under 18 a parent does (FR-2.1, NFR-4.4).
- At least 10 lessons in English, 5 or more of them with mixed English and Russian speech, and at least 10 in Russian.
- Some recordings have background noise, and some have a poor microphone.
- For every recording a reference text is written by hand (chosen fragments, at least 10 minutes) and the speakers are marked.
- For 5 to 10 recordings the teacher uploads the materials in advance and marks, topic by topic, whether the student answered correctly. This is the control set (FR-16.2).
- 3 to 5 more Kazakh lessons, used only for the control WER and not for acceptance.

Subset	Recordings	Used for
Development	5, in addition to the test set (proposed)	Tuning prompts, thresholds and settings
Test	20 or more: at least 10 English (5 or more mixed), at least 10 Russian, noisy and poor-microphone ones included	Final MVP metrics NFR-5.1 to NFR-5.4 and NFR-5.6; frozen, never used for tuning
Control set	5 to 10 lessons with the teacher's rating of each topic (FR-16.2)	NFR-5.5, index variants (C3), comparison of models (C7); never used for training
Kazakh control	3 to 5	Control WER only
Training set (stage 2)	Teachers' confirmations and changes from the pilot (FR-8.7, FR-16.1), only anonymized and only with the consent of FR-2.3; at least 300 examples, goal 500	Fine-tuning (FR-16.4)
Open data (trial)	MathDial, NCTE Transcripts; licences checked first	Trial fine-tuning (FR-16.0)

A few rules hold for all of this data. The development and test sets never share a lesson, and if possible the same teacher and student pair stays out of one of them. Training and control examples are split by lessons, not by fragments (FR-16.3), so no control lesson ever reaches training. With fewer than 300 examples the model is still trained on what is available, and this is stated in the thesis. Before rating the control lessons, the teacher gets a short written instruction about what a topic is and what each status means. The data is personal: it lives only on the team server in Kazakhstan (NFR-4.1), never in Git, and is deleted on the terms of the consent. Results in the thesis are anonymized.

6.2 Environment

Item	Value
GPU server	NVIDIA GPU with at least 16 GB (at least 24 GB with a local LLM); 8 CPU cores, 32 GB RAM, SSD 500 GB. The exact GPU is written in every results table

Item	Value
GPU for fine-tuning	At least 24 GB for QLoRA of a 7 to 9 billion model; a university server or a rented GPU in Kazakhstan; the data is deleted after training (NFR-4.5)
OS and drivers	Ubuntu 22.04 or 24.04 LTS, NVIDIA drivers and CUDA; versions recorded
Deployment	All services in Docker Compose (NFR-7.1), the same images for every run
Versions	Git commit or tag, model names and versions, prompt version (FR-15.3), training data and settings (NFR-7.7), library versions: saved with every result
LLM settings	Fixed temperature 0.2; each LLM test runs 3 times to see the spread
Load tests	30 virtual users (k6); test database with generated data, no real personal data
Frontend build	Production build (Vite) served by nginx in Docker Compose, as on the pilot server; commit recorded
Screen sizes	360 × 800 (phone, NFR-6.2), 768 × 1024 (tablet), 1440 × 1024 (desktop, size of the mockups)
Browsers	Two last versions of Chrome; also Edge, Firefox, Safari (WebKit in Playwright, a real iPhone or Mac if available); versions recorded
Frontend test data	Mock API (Mock Service Worker) with generated data for daily tests; real API with the test database for S1 to S9
Fallback	Without the GPU server, research runs in Colab or Kaggle, but only on open data and on recordings whose participants allowed processing abroad. Real lessons are processed only in Kazakhstan. Final timing is measured only on the target server

6.3 How numbers are reported

The dataset is small, and the control set has only 5 to 10 lessons. So every quality metric gets a 95 % confidence interval (bootstrap over recordings, 1,000 samples). The results are also split by subset: English, Russian or mixed; clean, noisy or poor microphone. This shows where the system is weaker.

7. End-to-end scenarios

S1 to S9 belong to the MVP, S10 to S12 to stage 2. Playwright runs each scenario where it can, and the result (pass or fail, date, commit) goes into the results table.

#	Scenario	Requirements	Pass criterion
S1	A tutor registers an organization, adds a student and invites a parent; the parent gives consent	FR-1, FR-2.1, FR-2.2	Consent saved with who, when and the text version; upload is blocked before consent
S2	The teacher uploads a PDF with materials; the LLM proposes topics; the teacher renames one, merges two and confirms the list	FR-3.4 to FR-3.6	Only confirmed topics are used in the next analysis

#	Scenario	Requirements	Pass criterion
S3	The teacher uploads a 60-minute video of an English lesson; the status goes through all stages	FR-4, FR-5, FR-6, FR-7, NFR-1.1	Report ready in 30 min or less; the video file is deleted after the audio is taken out (FR-4.4)
S4	Speaker labels are right, also in the Russian parts; the teacher fixes a wrong label	FR-6.3, FR-6.4	The analysis runs again and the report is updated
S5	The teacher changes the status of one topic and confirms another	FR-8.7, FR-8.6, FR-16.1	Index and knowledge map are recalculated; the change is in the history and saved as an example
S6	Teacher, student and parent each open their own report; a click on a time code opens that moment	FR-9.1 to FR-9.4, FR-11 to FR-13	Different texts in the language of each profile; the player jumps to the time code; the knowledge map is updated
S7	A parent of another student tries to open someone else's report in the interface and by a direct API request	FR-1.4, NFR-3.4	HTTP 403 or 404, no data returned
S8	30 days after upload the audio is deleted (tested with a shorter period)	FR-4.7, NFR-4.3	File gone from storage; transcript and report remain; the time code opens the transcript
S9	A parent withdraws consent; the system administrator deletes the data	FR-2.4	Recordings, transcripts and reports of the student are gone within 3 working days
S10	(Stage 2) The student gets a test on topics with an index below 80 % and passes it	FR-10.4, FR-10.7, FR-10.8	Index changes by $I = 0.4 \cdot D + 0.6 \cdot T$; colors on the map change
S11	(Stage 2) After the third lesson in a row with a low index, the topic becomes "stuck"; on Monday the parent and the teacher get the weekly summary	FR-8.4, FR-11.2, FR-9.5, FR-14.1	"Need attention" block, notification and summary appear
S12	(Stage 2) The system administrator switches the analysis to the fine-tuned model; a new lesson is processed by it	FR-7.7, FR-15.3, FR-16.6	The report shows the new model version

7.1 Frontend part of the scenarios

The Playwright test checks the following on screen. The screen names are the names of the Figma frames.

#	Screens	What is checked in the interface
S1	Registration, Invitation, Consent	Form checks and error messages; the parent sees the consent text, the required checkbox and a separate optional checkbox for model training (FR-2.3); upload for this student is blocked until consent
S2	Materials, Topic list	Format and size are checked before upload; the topic list appears after processing; rename, merge, delete and add work; only confirmed topics are marked as active

#	Screens	What is checked in the interface
S3	Upload a recording, Lessons and statuses	Language choice; file check before upload; progress bar; the stages appear one by one without reload; after an error the "Restart" button works (FR-4.6)
S4	Transcript and speakers	Roles are shown in color (FR-6.7); the teacher changes a label; the page warns that the analysis will run again; the status page shows the new run
S5	Teacher's report, Knowledge map	"Confirm" and "Change" on each topic, comment is optional; the new color and index appear without reload; parts outside the materials have an "Add topic" action (FR-7.8)
S6	Reports for teacher, student and parent	Three different texts for one lesson in each user's report language (FR-9.2); a click on a time code moves the player to that second (± 1 s); the parent sees no player in the MVP (FR-13.4 is stage 2)
S7	Parent cabinet, child switch	Only own children are in the switch (FR-13.1); a direct link to another student's report shows the "No access" page and no data; the network log shows 403 or 404
S8	Lesson report after 30 days	The player is replaced by the transcript fragment; time codes open the fragment (FR-9.4)
S9	Student card (system administrator)	After deletion, the student's lessons and reports are gone from all three cabinets

Pages with a phone version (parent home, student reports, teacher statuses) also go through every scenario at 360 × 800.

8. Schedule

Period	Stage	Verification work	Claims
Oct to Nov 2026	Preparation, approval of requirements	Collect at least 20 recordings (English and Russian) and 3 to 5 Kazakh ones with consent; reference texts and speaker labels; CI with unit tests	None
Oct to Nov 2026	Frontend: mockups	Figma mockups of all cabinets; usability test of the prototype (B9); frontend CI with ESLint, TypeScript, Vitest	C6
Nov 2026	Research	First WER on real lessons, mixed speech first	C1
Nov to Dec 2026	Model research	Cleaning comparison (ΔWER), WER by language, DER, control WER on Kazakh, baselines B1 to B3	C1
Nov 2026 to Feb 2027	Backend and frontend	Unit and integration tests in CI, coverage 50 % or more; login, roles, student list, materials, upload and statuses on the mock API; first Playwright tests (S1 to S3)	C5
Dec 2026 to Feb 2027	LLM analysis and index	Comparison of cloud LLMs (B6) on the control set; agreement, kappa, B4, B5; JSON rate; index variants (B7)	C2, C3
Jan to Feb 2027	Frontend: reports	Reports, player, knowledge map, "Confirm" and "Change"; Lighthouse and bundle size; time-code accuracy	C4, C5

Period	Stage	Verification work	Claims
Late winter 2027	Trial fine-tuning	QLoRA on MathDial or NCTE, comparison with the base model (FR-16.0)	C7
Feb to early Mar 2027	Integration, MVP	Real API instead of the mock; S1 to S9 in all browsers; load, reliability and security tests; phone layout, accessibility and the metrics of section 4.4	C4, C5
Before the pilot	None	Manual process (B8): teacher time diary for one week	C6
Mar to Apr 2027	Pilot and stage 2	2 to 3 teachers for 2 to 4 weeks: share of unchanged statuses, time diary, questionnaires, SUS; scenarios S10 and S11; fixes in short releases	C6
Early spring 2027	Fine-tuning on corrections	At least 300 examples (goal 500); split by lessons; comparison of three models (B10); scenario S12	C7
Apr to May 2027	Explanatory note	Final results tables	All

9. Results and reporting

Results tables and charts go to docs/results/, and the evaluation scripts to src/ml_service/eval/ and tests/, so anyone can repeat every number in the thesis. Frontend results go to docs/results/frontend/: Lighthouse reports, the Playwright HTML report with screenshots at 360, 768 and 1440 px, the axe-core report and the bundle size report. The Playwright tests live in src/web/e2e/. Every result row has the date, the commit or tag, the model and prompt version, and the subset.

Results table for C1:

Setup	Subset	WER (95 % CI)	DER (95 % CI)	Time per 60 min
B1: no cleaning	noisy			
Final: cleaning + large-v3	noisy			
Final: cleaning + large-v3	English			
Final: cleaning + large-v3	Russian			
Final: cleaning + large-v3	mixed			
Final: cleaning + large-v3	Kazakh (control)			

Results table for C2 and C7:

Model	Transcript	Status agreement (95 % CI)	Valid JSON	Time and cost per lesson
Cloud LLM (MVP)	checked			
Cloud LLM (MVP)	recognized			
Base open model	recognized			

Model	Transcript	Status agreement (95 % CI)	Valid JSON	Time and cost per lesson
Fine-tuned open model	recognized			

Results table for the frontend (section 4.4):

Page	Device	LCP, s (median of 3)	Steps to main action	axe-core critical
Teacher home	desktop			
Lesson report	desktop			
Parent home	phone, 360 px			

The thesis will report the cleaning comparison by WER; WER for English, Russian and mixed speech, DER and teacher detection; the control WER for Kazakh; the comparison of 2 to 3 cloud LLMs by agreement with the teacher and valid answers; the analysis on the checked and on the recognized transcript; the trial and the main fine-tuning (base, fine-tuned and cloud model); and the pilot results.

10. Risks to valid results

Risk	Effect	What is done
Small test set (20 recordings) and small control set (5 to 10 lessons)	Wide uncertainty, mostly for mixed speech and for NFR-5.5	Confidence intervals; results by subset; more lessons if they can be obtained
Mixed speech is recognized worse than expected	NFR-5.1 for mixed speech is not reached	First WER in November; compare large-v3 and turbo on mixed speech; mark such lessons as lower accuracy
Tuning on the test set	Results look better than they are	Separate development set; the test set is frozen after the research stage; the control set is never used for training (FR-16.3)
One teacher rates the control lessons	The rating can be subjective	Written rating instruction; kappa is reported and the topics where the system and the teacher disagree most are examined
Teachers press “Confirm” without reading	NFR-5.8 looks better than it is	Compare with the control set, where the teacher rates on their own; look at the time spent on a report in the logs
Few examples for fine-tuning	The fine-tuned model is not better than the base one	Trial on open data first; examples are collected from the first pilot lesson; if there are fewer than 300, this is stated in the thesis
The fine-tuned model does not reach the cloud model	No switch to a model of the project’s own	The cloud LLM stays; the comparison is a research result
LLM answers change between runs	Results cannot be repeated	Temperature 0.2, 3 runs, fixed prompt version

Risk	Effect	What is done
No GPU server	Real lessons cannot be processed	Agree on a university server or a rented GPU in Kazakhstan in early November; until then only open data in Colab or Kaggle
Not enough recordings with consent	The dataset is late	Start with the teachers already interviewed; have the consent text ready before the first upload
Few pilot users	Pilot results are not general	They are presented as a case study, not as proof for all schools
The API is not ready when the frontend needs it	Frontend tests wait for the backend	Work from the OpenAPI file and a mock API; the same Playwright tests later run on the real API
Mock data is too clean	Pages look fine in tests but break with real data (long names, empty lists, errors)	Mock data has long texts, empty states and error answers; final checks only on the real API with real lessons
Page load measured on a fast computer	NFR-1.4 looks better than it is	Fixed throttling (10 Mbit/s, CPU 4 times slower in Lighthouse), production build, empty cache
Winter session and holidays during the frontend work	Less time for tests	Playwright tests are written together with the pages; S1 to S9 must pass before the MVP in early March